

Clutch Nexus

Supply Chain Disruption Assistant & Fleet Utilisation Optimizer

IBM Bob Hackathon 2026

Track: AI

Team: Clutch Nexus

Team Members

Bhargav Zalariya – Team Lead

Prey Patel

Himesh Male

Kanan Sharma

From disruption detection to recovery – one connected supply-chain control tower.

Supply-Chain Disruptions Create a Chain Reaction

A disruption rarely affects only one shipment. Traditional response is fragmented – each layer of impact handled in isolation, slowing recovery and compounding risk.

The Cascade

01

Disruption

02

Affected Shipments

03

Routes & Network Nodes

04

Fleet Capacity

05

Cold-Chain Risk

06

Financial Exposure

07

Recovery Decisions

Key Challenges

Disruptions can affect multiple shipments and routes simultaneously

Operations teams need to identify the full downstream impact quickly

Fleet capacity may become poorly utilised while urgent cargo is delayed

Temperature-sensitive cargo requires additional monitoring

Recovery options involve trade-offs between time, cost, capacity and risk

Decision-making across these factors is often fragmented



When a disruption occurs, what is affected, how severe is the impact, and what recovery action should be taken?

Clutch Nexus: A Connected Supply-Chain Recovery Control Tower

Clutch Nexus connects disruption analysis and operational recovery in a single workflow.



Detect

Identify active supply-chain disruptions



Cascade

Trace impact across shipments, routes, network nodes, fleet capacity, cold-chain cargo, and financial exposure



Prioritise

Identify critical shipments and high-risk cargo



Optimise

Use Google OR-Tools CP-SAT to generate fleet redeployment recommendations



Simulate

Compare alternative recovery strategies using What-If scenarios



Monitor

Track cold-chain excursions and network conditions



Impact Cascade Engine – connects a disruption to its operational and financial consequences instead of treating it as an isolated alert.

How Clutch Nexus Works

System Architecture

Next.js Frontend

Dashboard · Maps · UI

FastAPI Backend

REST API layer – Python 3.11+

Core Engines

Impact Cascade Engine · Fleet Optimizer (OR-Tools CP-SAT) · What-If Scenarios

Operational Data Layer

SQLite + SQLAlchemy · Pydantic v2

Visualisation

Recharts · Leaflet · Cold-Chain Telemetry + Network Data

Technology Stack

Layer	Technology
Frontend	Next.js 14, TypeScript, Tailwind CSS
Backend	FastAPI, Python 3.11+
Optimisation	Google OR-Tools CP-SAT
Data	SQLite, SQLAlchemy, Pydantic v2
Visualisation	Recharts, Leaflet
Development	IBM Bob

One Disruption. Complete Operational Visibility.

Impact Cascade Engine

Maps the full chain reaction: Disruption → Shipments → Routes → Fleet → Cold Chain → Financial Impact

Fleet Optimisation

Google OR-Tools CP-SAT evaluates available capacity and constraints to generate fleet redeployment recommendations.

Cold-Chain Monitoring

Identifies simulated temperature excursions for temperature-sensitive shipments.

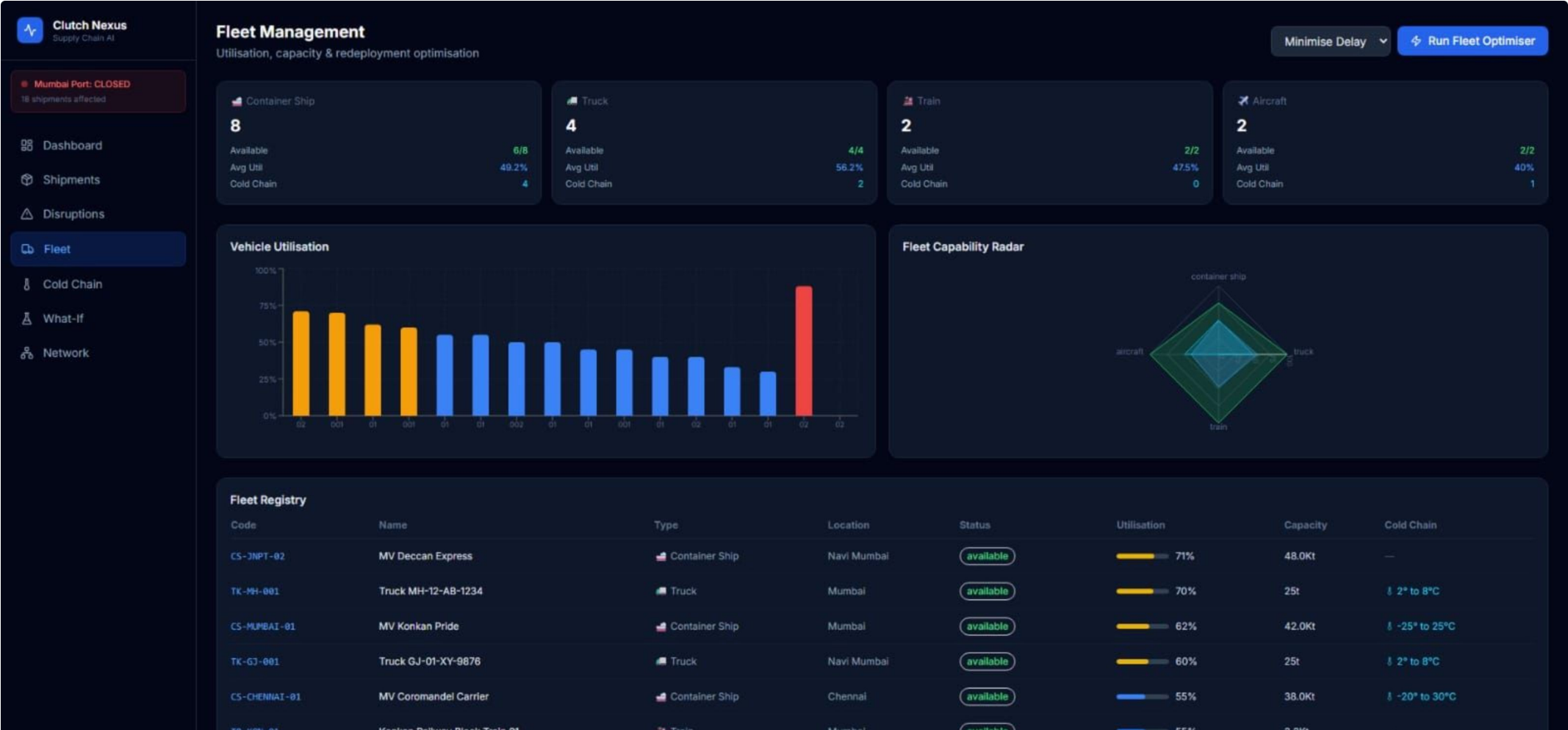
What-If Simulator

Tests alternative recovery strategies and compares delay, shipments recovered, risk reduction, and additional recovery cost.

Multimodal Network

Interactive Leaflet map showing network nodes, connections and disruption locations.

Bottom line: From "What happened?" to "What should we do next?"



Built with IBM Bob + AI-Assisted Engineering

IBM Bob

IBM Bob was used throughout development to accelerate application architecture, frontend and backend implementation, feature development, testing and debugging, documentation, and deployment preparation.

AI-Assisted Development

The solution was developed through an AI-assisted software engineering workflow, while keeping the core operational calculations deterministic and explainable.

Core Intelligence

Impact Cascade Engine

Deterministic dependency mapping · Shipment impact analysis · Network propagation · Financial exposure calculation

OR-Tools CP-SAT

Constraint-based fleet optimisation · Capacity allocation · Priority-aware recovery · Operational recommendations



The innovation is not just detecting a disruption – it is connecting the disruption to its cascading operational impact and recovery decisions.

From Disruption to Recovery Decisions

Prototype Dataset

30

Shipments

16

Network Nodes

3

Active Disruptions

52.9%

Baseline Fleet Utilisation

\$78.8M

Cargo Value at Risk

\$127M

Estimated Total Impact

Demonstrated Recovery Scenario: Mumbai Port Partial Closure

Metric	Baseline	JNPT Reroute
Average Delay	108h	36h
Shipments Recovered	–	12
Risk Reduction	–	42% lower
Additional Cost	–	+\$1.2M



Key Message: The What-If engine makes recovery trade-offs visible before an operational decision is made.

All figures shown are from a deterministic synthetic prototype dataset and are not live operational/customer data.

Clutch Nexus

Our Vision

Build a connected control tower that helps supply-chain teams move from disruption detection to coordinated recovery decisions.

Detect Supply-chain disruptions	Understand Cascading operational impact	Optimise Fleet recovery with OR-Tools
Protect Cold-chain shipments	Compare Recovery strategies through What-If simulation	

Built with IBM Bob

Bhargav Zalariya · Prey Patel · Himesh Male · Kanan Sharma

When disruption happens, Clutch Nexus shows what breaks – and what can be recovered.